

Overtune 3 (ot3) — System Installation Guide

NATIVE INSTALLER, IN-APP AUTO-UPDATER, PORTABLE ZIP, AND SOURCE BUILD

1. System Overview & Hardware Requirements

Overtune 3 (ot3) is a high-precision digital filter design, audio simulation, and DSP study workstation built on modern C++20 and Qt 6. It features automated mathematical audits, sub-millidecibel magnitude precision, SIMD cascaded biquad evaluation, and interactive DSP Chat assistance.

- **Windows Requirements:** Windows 10 (64-bit, 1809+) or Windows 11 (x64) with Microsoft Visual C++ 2015–2022 Redistributable.
- **Linux Requirements:** Any modern 64-bit Linux distribution (Ubuntu 22.04+, Fedora 38+, Arch Linux, Debian 12+) with GLIBC 2.34+ and Qt 6.4+ libraries.
- **Audio & Display:** DirectSound or WASAPI audio output device; minimum 1280×800 resolution with HiDPI support.

2. Windows Installation & Deployment

Overtune 3 provides four convenient installation and execution options on Windows:

Method A: Standalone Native Installer (ot3-installer.exe)

The dedicated graphical setup wizard handles complete deployment with no admin rights required:

- Run `ot3-installer.exe` (or `Overtune3-Setup-x64.exe`) from `dist/`.
- Configure target destination (defaults to `%LOCALAPPDATA%\Programs\Overtune3`).
- Automatically creates Start Menu and Desktop shortcuts and registers in Windows **Installed Apps**.

Method B: Standalone Portable ZIP (Zero Installation)

Extract `overtune3-windows-x64.zip` to any folder. Run `run.bat` or `bin\ot3.exe` directly. All Qt 6 libraries, QML modules, codecs, and fonts are self-contained.

Method C: Automated PowerShell Setup (install_windows.ps1)

For automated, unattended, or CI/CD deployments:

```
powershell -ExecutionPolicy Bypass -File .\scripts\install_windows.ps1
```

Optional flags: -InstallDir "C:\Path", -NoDesktop, -NoStartMenu.

Method D: Compiling from Source

```
build.bat          # Configures CMake, compiles Release, deploys Qt
    runtime
run.bat            # Runs the newly compiled release binary
```

3. Linux Installation & Execution

Method A: Native Automated Setup (install_linux.sh)

```
./scripts/install_linux.sh          # User-level (~/.local/share/overtune3)
sudo ./scripts/install_linux.sh     # System-wide (/opt/overtune3)
```

Method B: Standalone Portable Bundle

```
tar -xzf overtune3-linux-x86_64.tar.gz && cd overtune3-linux-x86_64 && ./ot3
.sh
```

4. In-App Updates, AI Assistant & Clean Uninstallation

- **In-App Updates:** Access **Settings** → **Updates** and click **Check**. Downloads are verified against release SHA-256 hashes.
- **AI DSP Assistant (Shadcy):** Configure OpenRouter model IDs and API keys in **Settings** → **AI Assistant**. On Windows, keys are encrypted securely using the Windows DPAPI.
- **Uninstallation:** Click **Uninstall** in **Settings** → **Resources & Maintenance**, via Windows **Settings** → **Apps** → **Installed apps**, or execute: %LOCALAPPDATA%\Programs\Overtune3\uninstall.bat

5. Command-Line Options & Essential Shortcuts

Option / Key	Type	Functionality
ot3 --install	CLI Flag	Launch directly into the native System Installation manager.
ot3 --update	CLI Flag	Query release server and display available version patches.
Ctrl + 1 .. 7	Keyboard	Navigate between Design, Analysis, Simulation, Export, Docs, Settings, Chat.
Ctrl + 0	Keyboard	Auto-scale and reset frequency, pole-zero, and impulse response plots.
F5 / Space	Keyboard	Start or pause the live real-time audio and DSP simulation engine.
Ctrl + T	Keyboard	Toggle between OLED Dark and Studio Light visual themes.

For full DSP derivations, refer to *theory_and_math.pdf* or open the in-app documentation browser.